16 Two vessels X and Y contain ideal gases at the same temperature T. The pressures of ideal gases in X and Y are P and $P/4$ respectively. The volume of X is 1.5 times that of Y. The vessels are connected by a narrow tube with a tap. The tap is initially closed. The temperature of the gas is maintained at the constant temperature T.

What is the pressure of the gas at equilibrium when the tap is opened?

A 0.50 P **B** 0.70 P **C** 0.75 P **D** 1.50 P